

VCagents, Composition as a First-Class Verification Problem.

Short form

VCagent (*n.*) - an agent that carries its composition obligations with it: it declares what it owns, observes, assumes, and guarantees, and it requires safety, progress, and mission success to be checked on the composed system rather than inferred from the correctness of its parts; other critical reflexes: may refuse deployment until the composition, not only its members, has been verified.

Formal definition

For agents k in a configured set C , a VCagent-oriented system makes the following obligations explicit:

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System == Glob /\ (\A k \in C : Spec(k))

System => Mission
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Here, **Glob** states system-level constraints; each **Spec**(k) identifies the agent's owned state and actions, environment behavior, assumptions, guarantees, and local progress conditions; and **Mission** states the required safety and liveness of the whole. Calling an agent a VCagent does not assert that these obligations have already been proved. It asserts that the agent exposes and preserves the structure needed to state, check, attribute, and recheck them.

Full definition

VCagent (*n.*; *from verification + composition + agent; pl. VCagents*) - an autonomous or semi-autonomous agent engineered to be specified, verified, and refined both as an individual component and as a member of a larger agentic system; whose main reflexes include declaring the state and actions it owns, the changes it may observe but does not cause, the behavior it attributes to other agents or the environment, and the global rules it leaves to the orchestrator; publishing explicit assumptions and guarantees so that interface compatibility and fault responsibility can be checked; and requiring safety, liveness, fairness, and mission satisfaction to be verified over the complete composition, especially when state is shared or actions are jointly enabled (*i.e., an agent that treats composition as a first-class verification object and carries its composition obligations with it; the compositional-verification counterpart of the autonomous agent*); other critical reflexes: recognizes that conjunction preserves stated safety constraints but does not by itself establish compatible assumptions, accountable ownership, non-vacuous progress, or mission success; searches the composed system for deadlock, circular waiting, Zeno behavior, and actions that are locally enabled but globally blocked; requires renewed evidence after material changes to membership, authority, interfaces, tools, assumptions, or operating conditions; and may refuse deployment until both its local proof obligations and the composition-level obligations are discharged.

Collective form

VCagents (*n., pl.; also used collectively*) - a system of VCagents whose configuration, interactions, and mission-level behavior are themselves explicit verification objects; not merely a collection of individually verified agents, but a composition for which ownership, contracts, global constraints, complete-system progress, and mission satisfaction are checked.